

Modular Anesthesia System

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Meet Our Team



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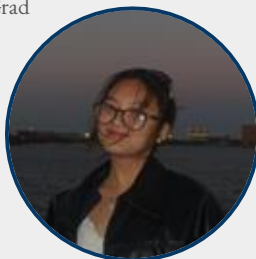
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Problem: Anesthesia setup is wasteful and time consuming

Anesthesiologists administer anywhere from ten to hundreds of doses of medications during each surgery and up to 10 surgeries per day. Some medications are “pushed” as discrete boluses while other medications are run continuously.

Current workflows require anesthesia providers to manually prepare each of these bolus medications prior to each surgery, requiring significant setup. Additionally, any unused medication is discarded at the end of each patient’s case.

Functional Requirements



Reduced volume of waste generated



Minimized setup time and medication change time



Variable dosing of medications with 0.1cc resolution



Simultaneous control of multiple medications



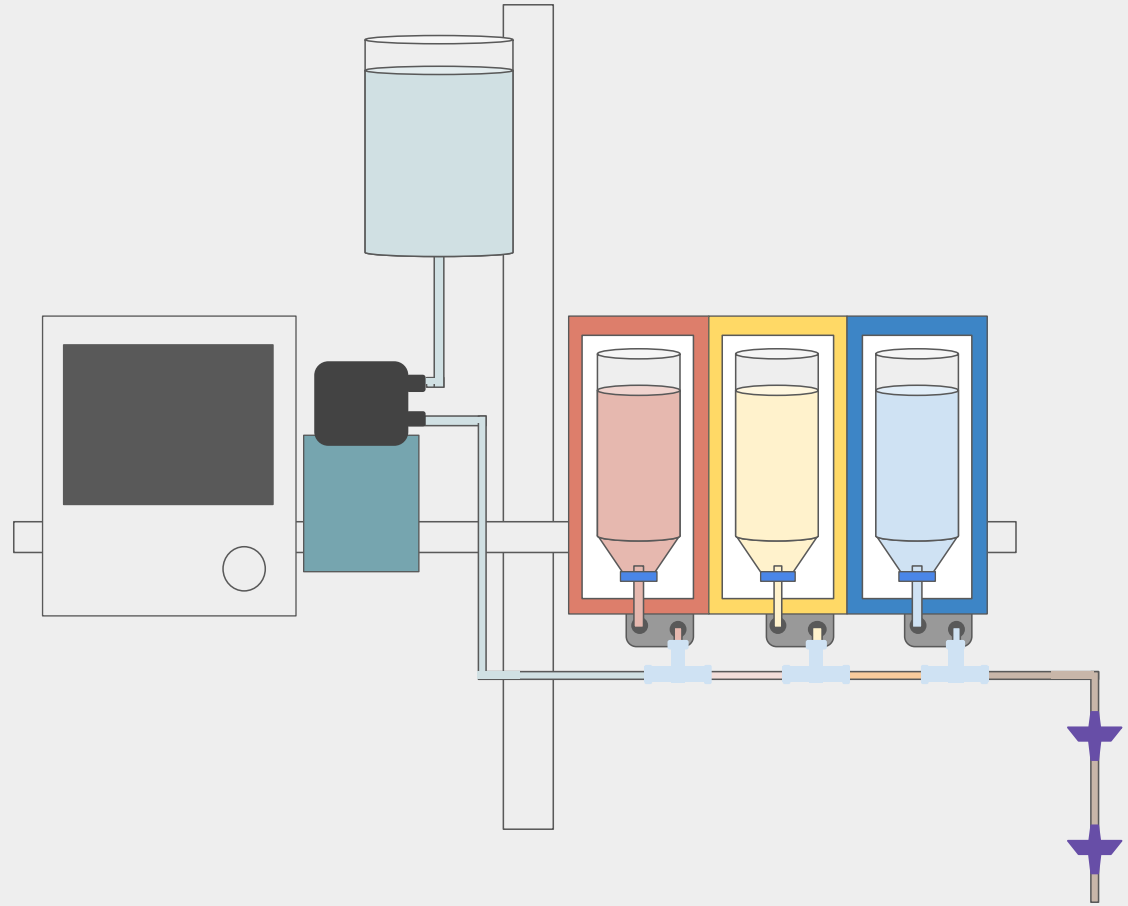
Meet 5 second medicine administration window



Compliance with relevant standards for safety/sterility

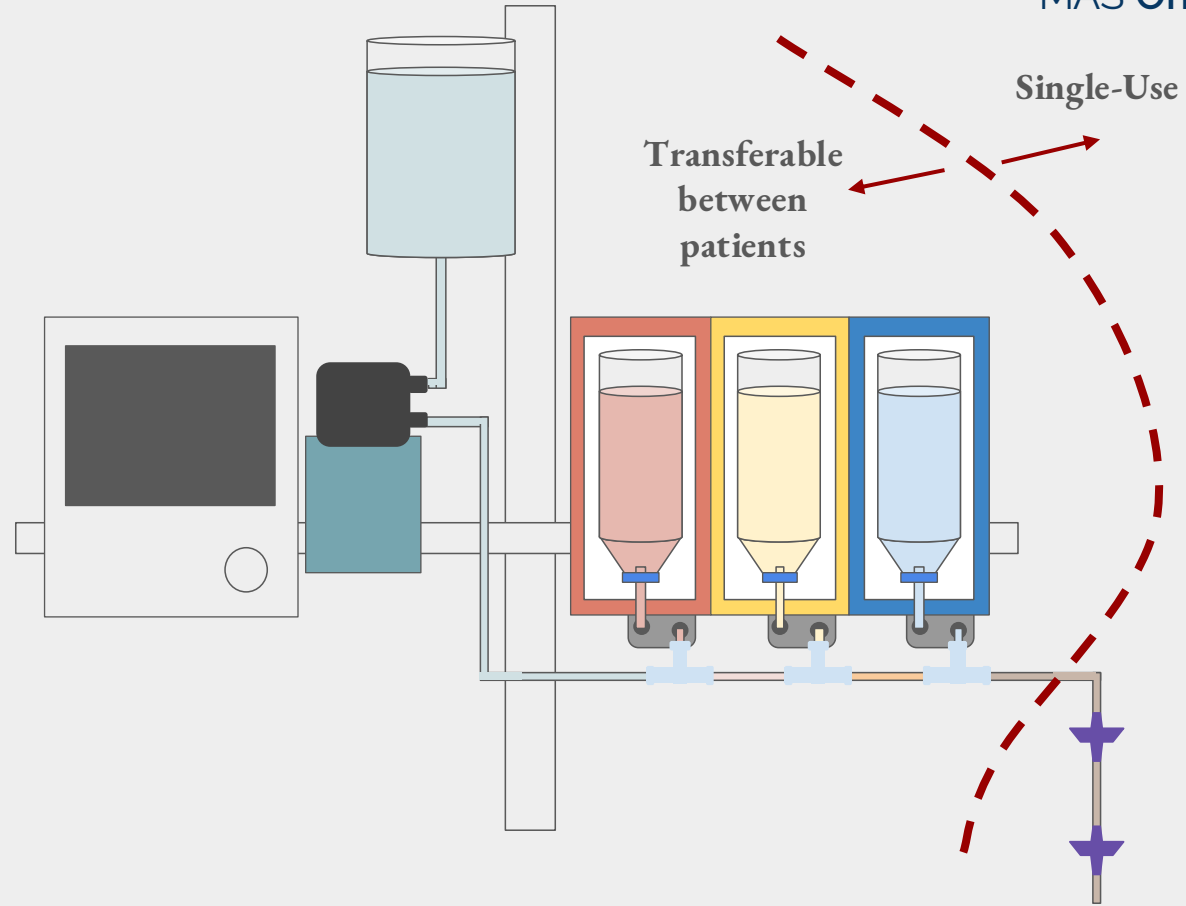
MAS One

A multi-medication
bolus dose
administration pump
that allows for a
transferable setup
between patients

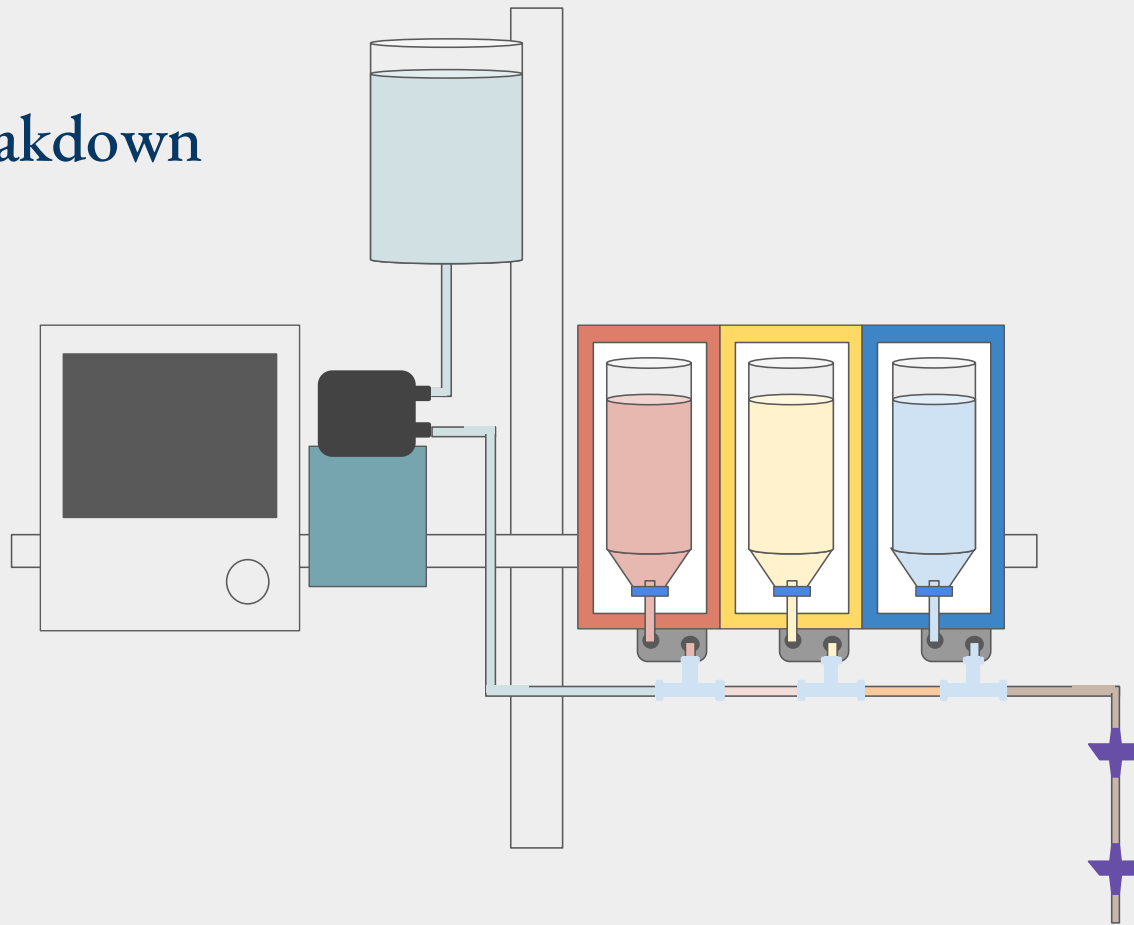


Key Features

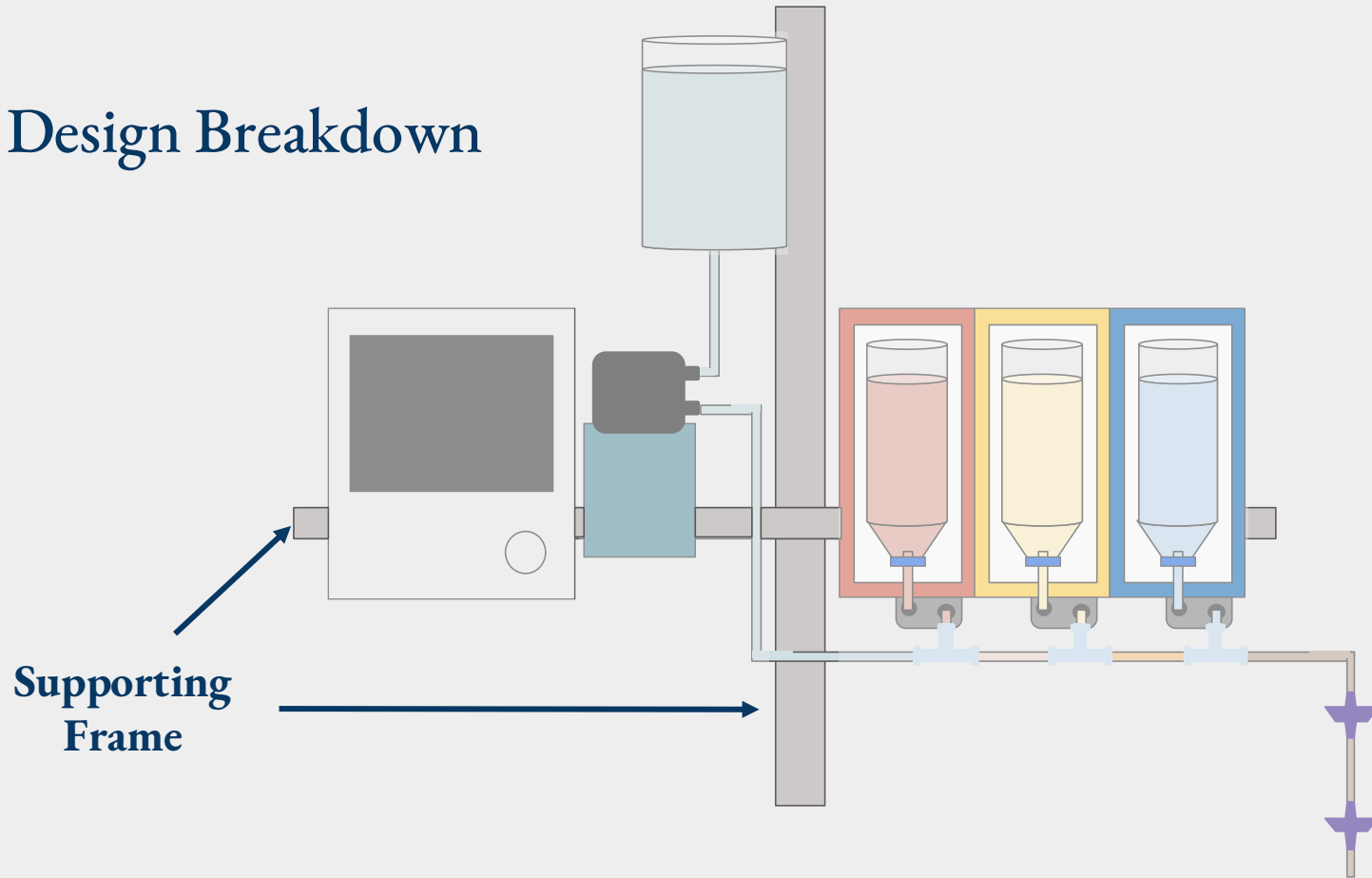
- Controls multiple pump modules from **one interface**
- **Eliminates single-use syringes**
- **Minimizes labor requirements** with “day use” components
- **Reduces medication waste**
- Prevents backflow through the IV allowing for **patient-to-patient transfers**



Design Breakdown

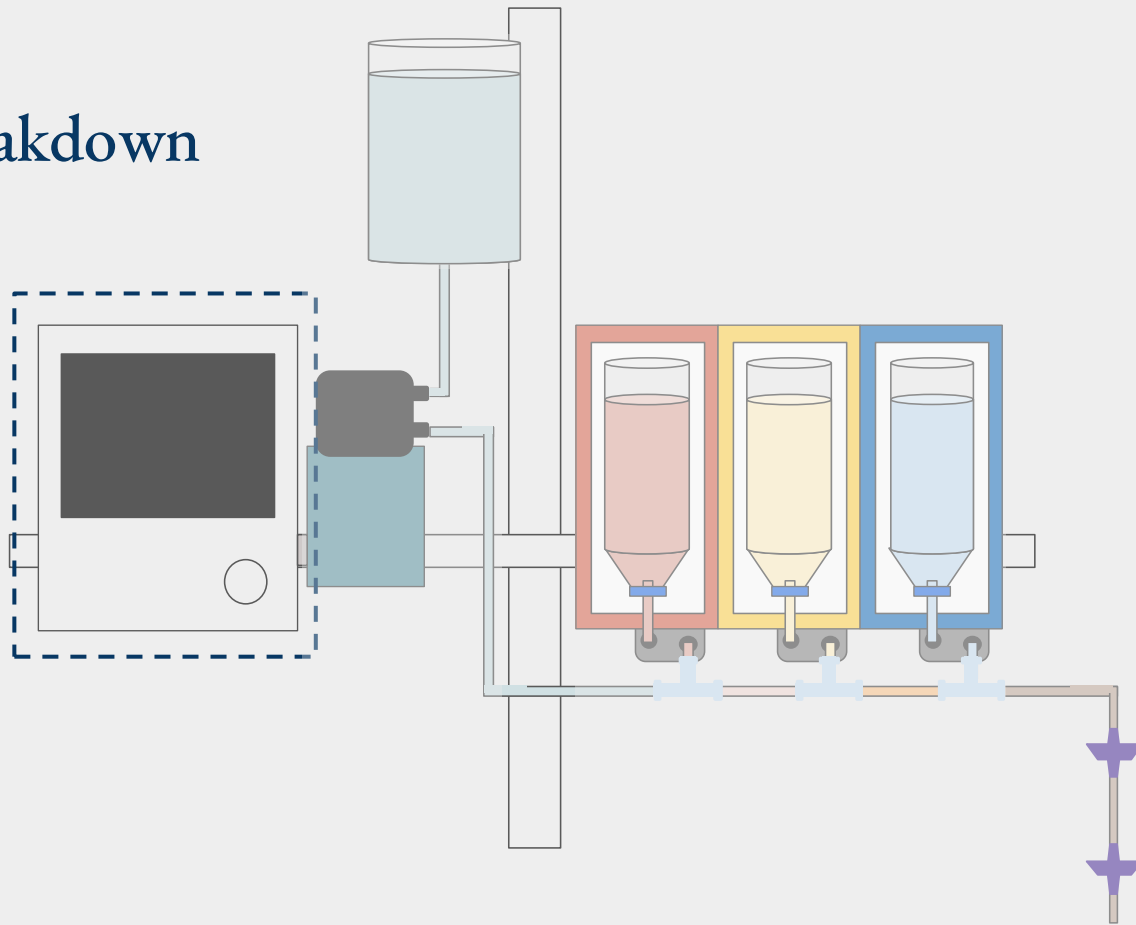


Design Breakdown

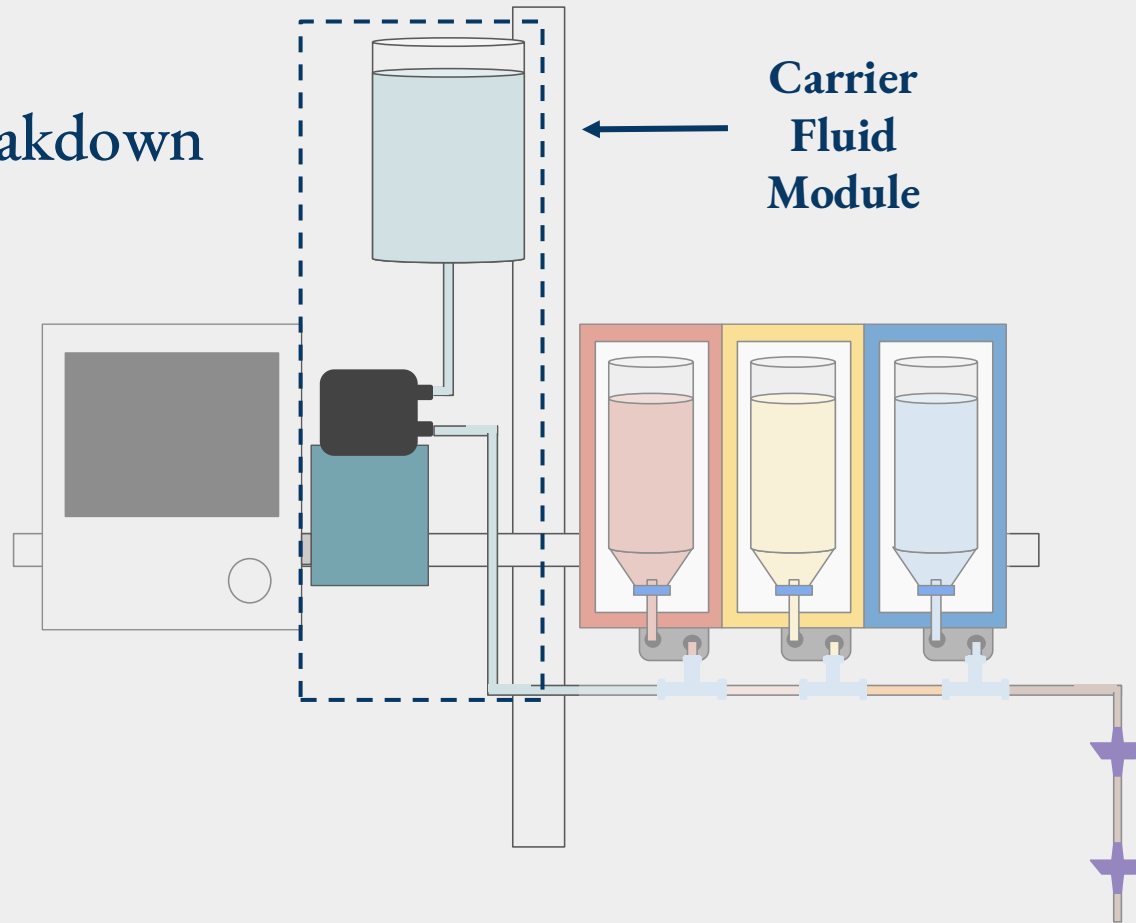


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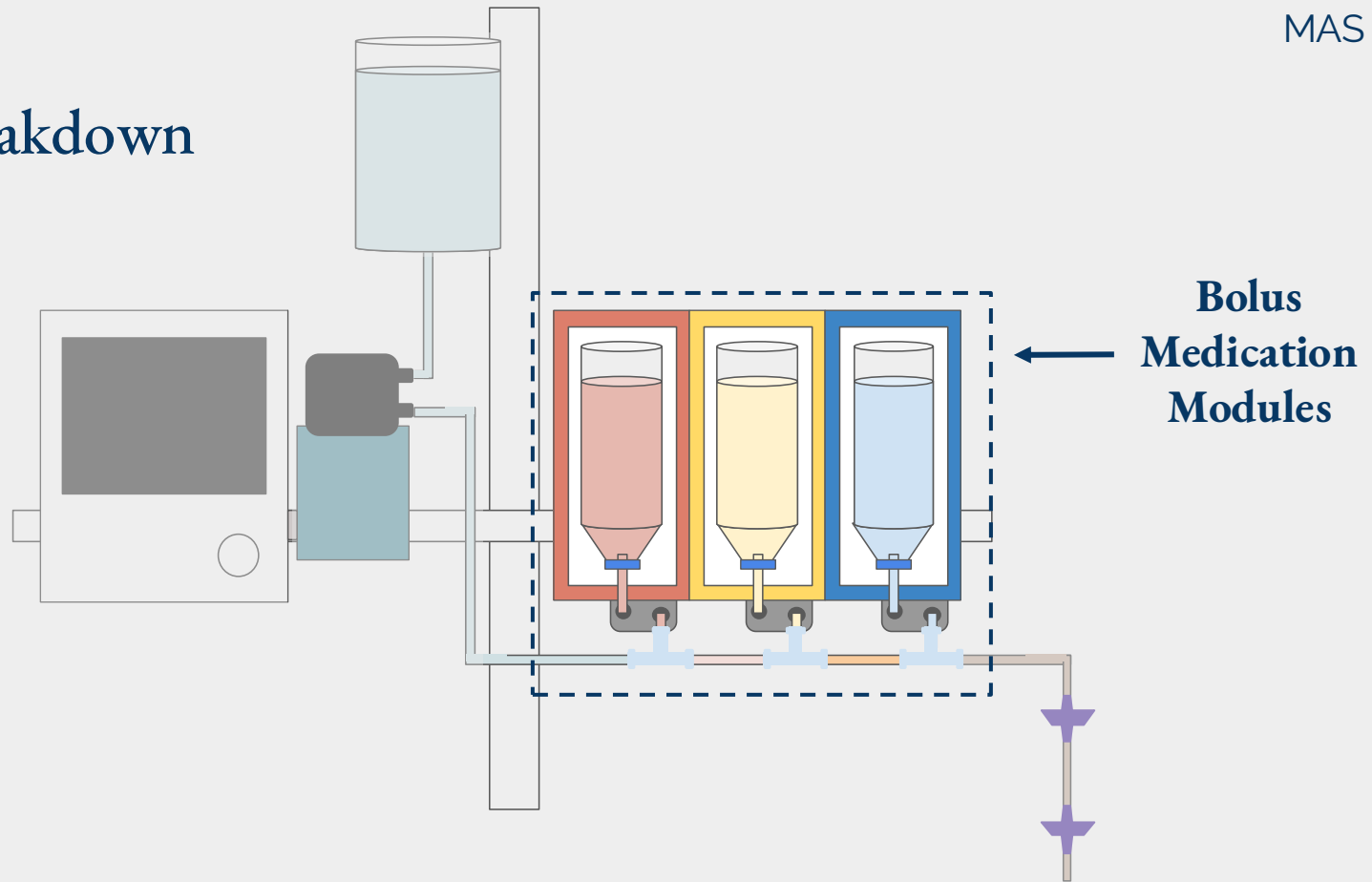
Central
Interface



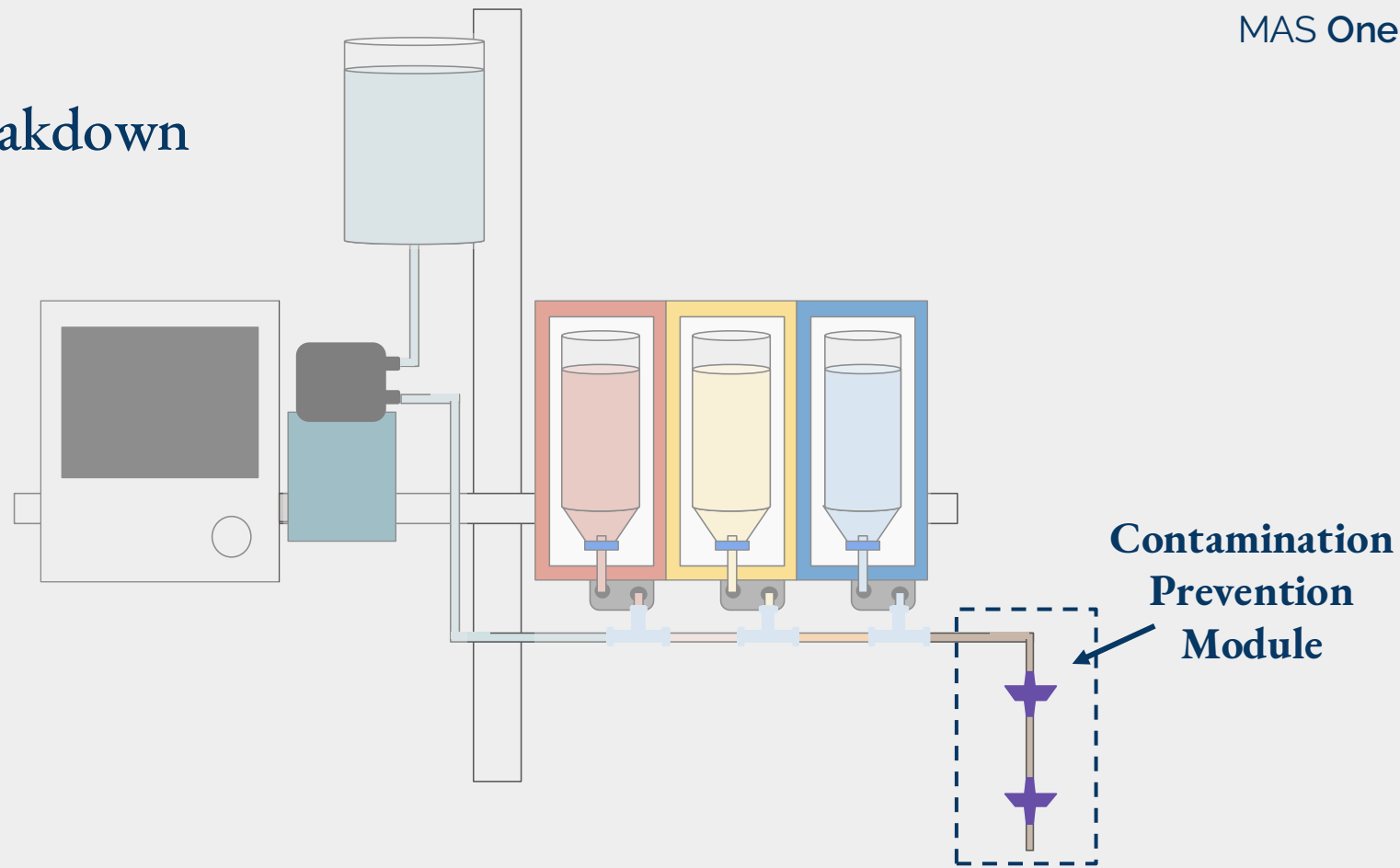
Design Breakdown



Design Breakdown



Design Breakdown



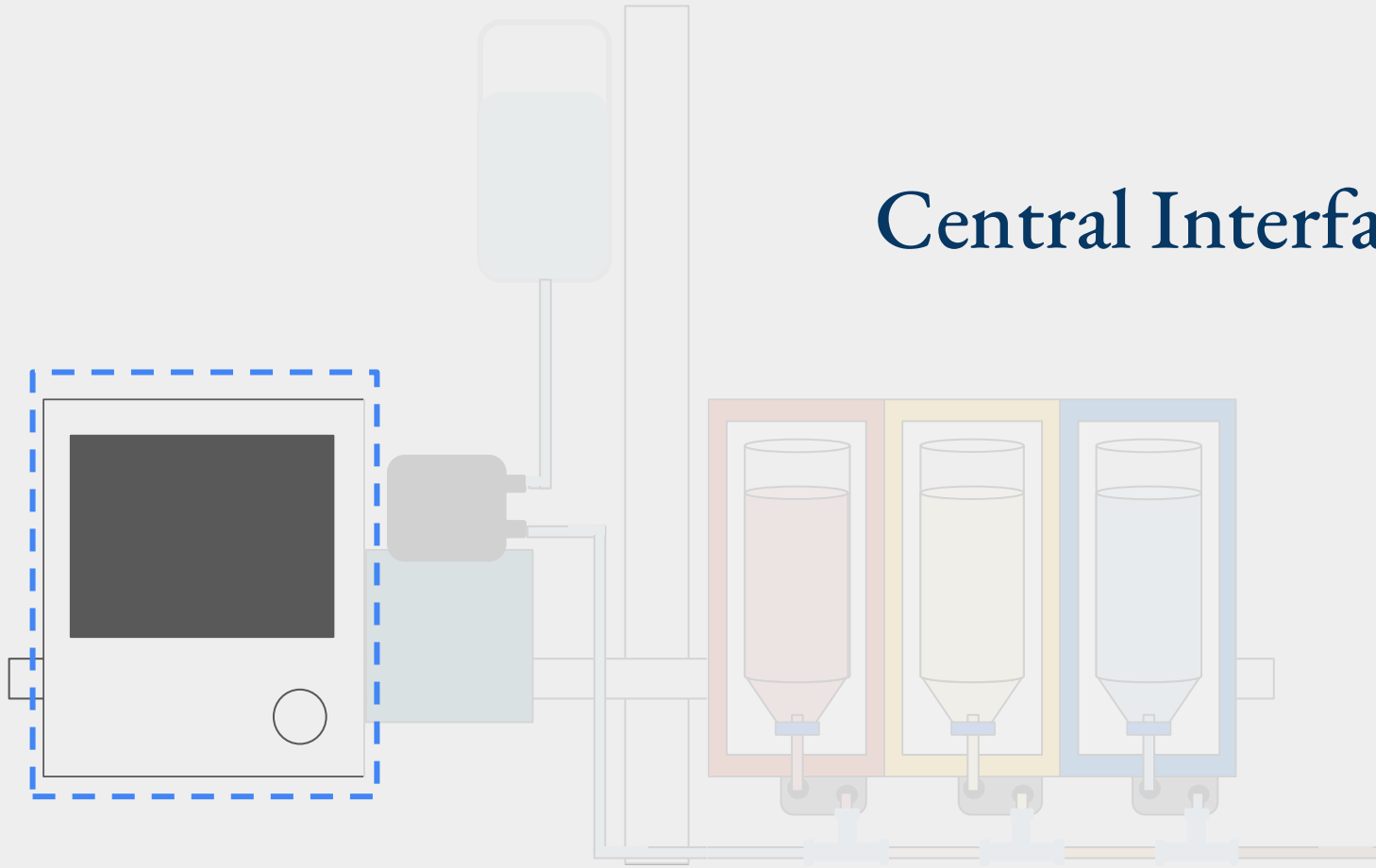
Design Details

Supporting Frame

The entire system attaches to an IV pole with a rail architecture for each module to slide on.

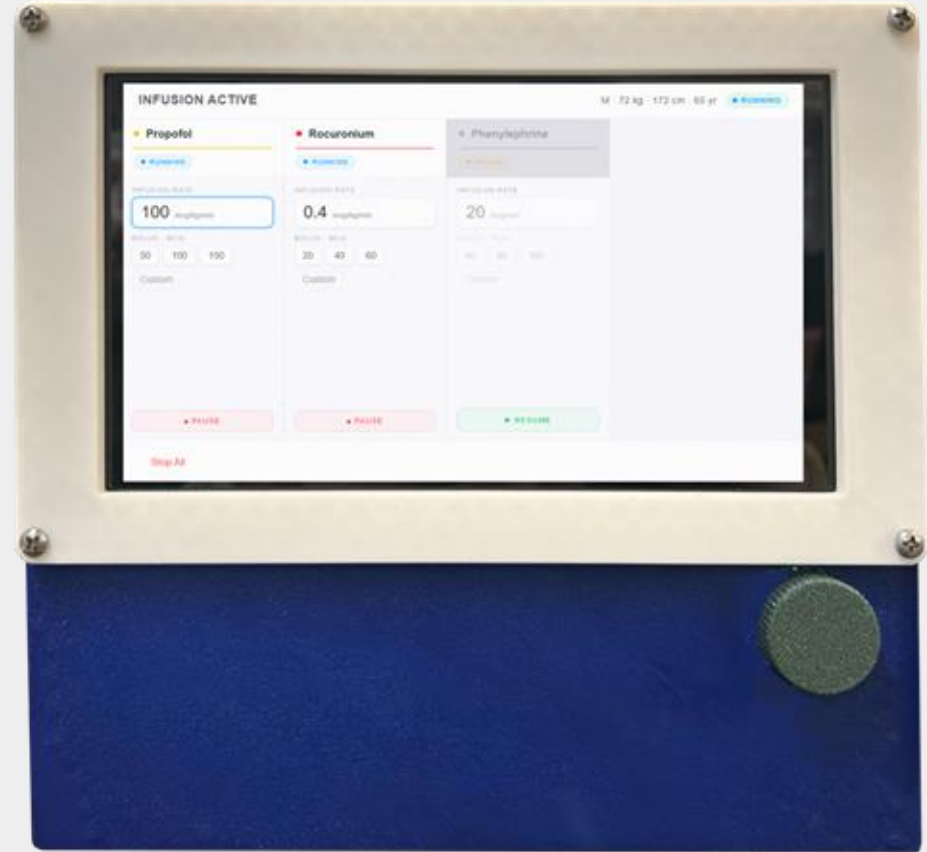


Central Interface



Central Interface - Mechanical Design

- Designed with 7" raspberry pi touchscreen and a single encoder for intuitive interfacing
- 3D printed housing that slides onto a supporting rail allowing for module re-configuration to user preference



PATIENT

SELECT RECORD

CONTINUE PRIOR

M

72 kg

172 cm

65 yr

LOAD FROM EHR

M

72 kg

172 cm

65 yr



New patient

NEW PATIENT

TAP A FIELD · ROTATE ENCODER TO ADJUST · PRESS TO CONFIRM & ADVANCE

WEIGHT

72

kg

HEIGHT

—

cm

AGE

—

yr

SEX

M

F

Back

Continue

INFUSIONS

M - 72 kg - 172 cm - 65 yr

HYPNOTIC

Propofol

10 mg/mL

☐

Etomidate

2 mg/mL

☐

Ketamine

10 mg/mL

☐

RELAXANT

Rocuronium

5 mg/mL

☐

Vecuronium

1 mg/mL

☐

Succinylcholine

20 mg/mL

☐

VASOPRESSOR

Phenylephrine

80 mcg/mL

☐

SELECTED

None selected

Back

Confirm

INFUSIONS

M · 72 kg · 172 cm · 65 yr

HYPNOTIC

Propofol

10 mg/mL



Etomidate

2 mg/mL



Ketamine

10 mg/mL



RELAXANT

Rocuronium

5 mg/mL



Vecuronium

1 mg/mL



Succinylcholine

20 mg/mL



VASOPRESSOR

Phenylephrine

80 mcg/mL



SELECTED

Propofol



Rocuronium



Phenylephrine



Back

Confirm

DOSE SETUP

M - 72 kg - 172 cm - 65 yr

● Propofol

10 mg/mL

INFUSION RATE

100 mcg/kg/min

BOLUS PRESETS - MCG

50

100

150

+ Add

● Rocuronium

5 mg/mL

INFUSION RATE

0.4 mcg/kg/min

BOLUS PRESETS - MCG

20

40

60

+ Add

● Phenylephrine

80 mcg/mL

INFUSION RATE

20 mcg/min

BOLUS PRESETS - MCG

40

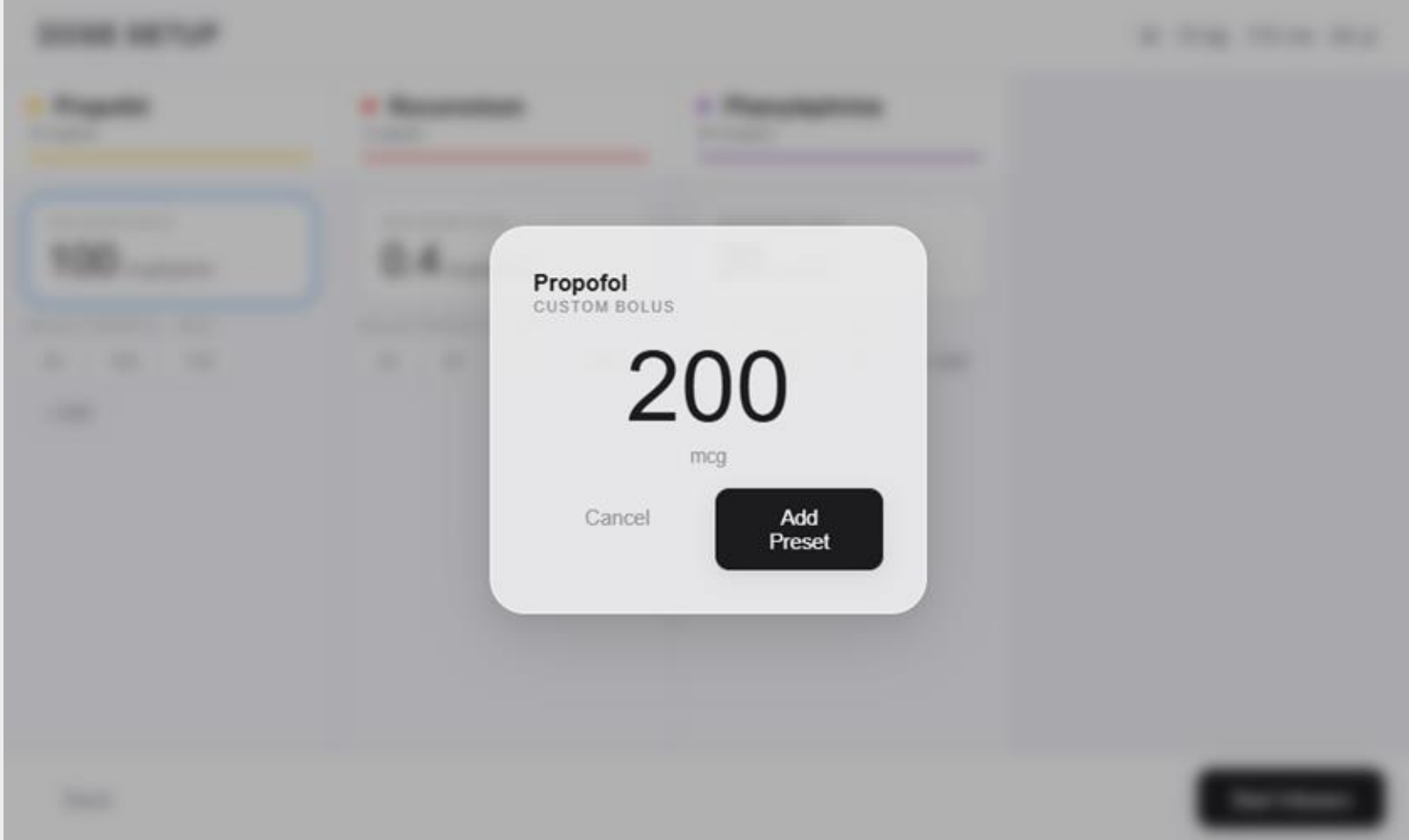
80

160

+ Add

Back

Start Infusion



Start Infusion?

Propofol: 100 mcg/kg/min

Rocuronium: 0.4 mcg/kg/min

Phenylephrine: 20 mcg/min

Cancel

Confirm

INFUSION ACTIVE

M · 72 kg · 172 cm · 65 yr

● RUNNING

● Propofol

● RUNNING

INFUSION RATE

100 mcg/kg/min

BOLUS · MCG

50

100

150

200

Custom

■ PAUSE

● Rocuronium

● RUNNING

INFUSION RATE

0.4 mcg/kg/min

BOLUS · MCG

20

40

60

Custom

■ PAUSE

● Phenylephrine

● RUNNING

INFUSION RATE

20 mcg/min

BOLUS · MCG

40

80

160

Custom

■ PAUSE

Stop All

Pause Rocuronium?

This will pause the Rocuronium infusion.
It can be resumed at any time.

Cancel

Confirm

INFUSION ACTIVE

M · 72 kg · 172 cm · 65 yr

● RUNNING

● Propofol

● RUNNING

INFUSION RATE

100 mcg/kg/min

BOLUS · MCG

50

100

150

200

Custom

■ PAUSE

● Rocuronium

● PAUSED

INFUSION RATE

0.4 mcg/kg/min

BOLUS · MCG

20

40

60

Custom

▶ RESUME

● Phenylephrine

● RUNNING

INFUSION RATE

20 mcg/min

BOLUS · MCG

40

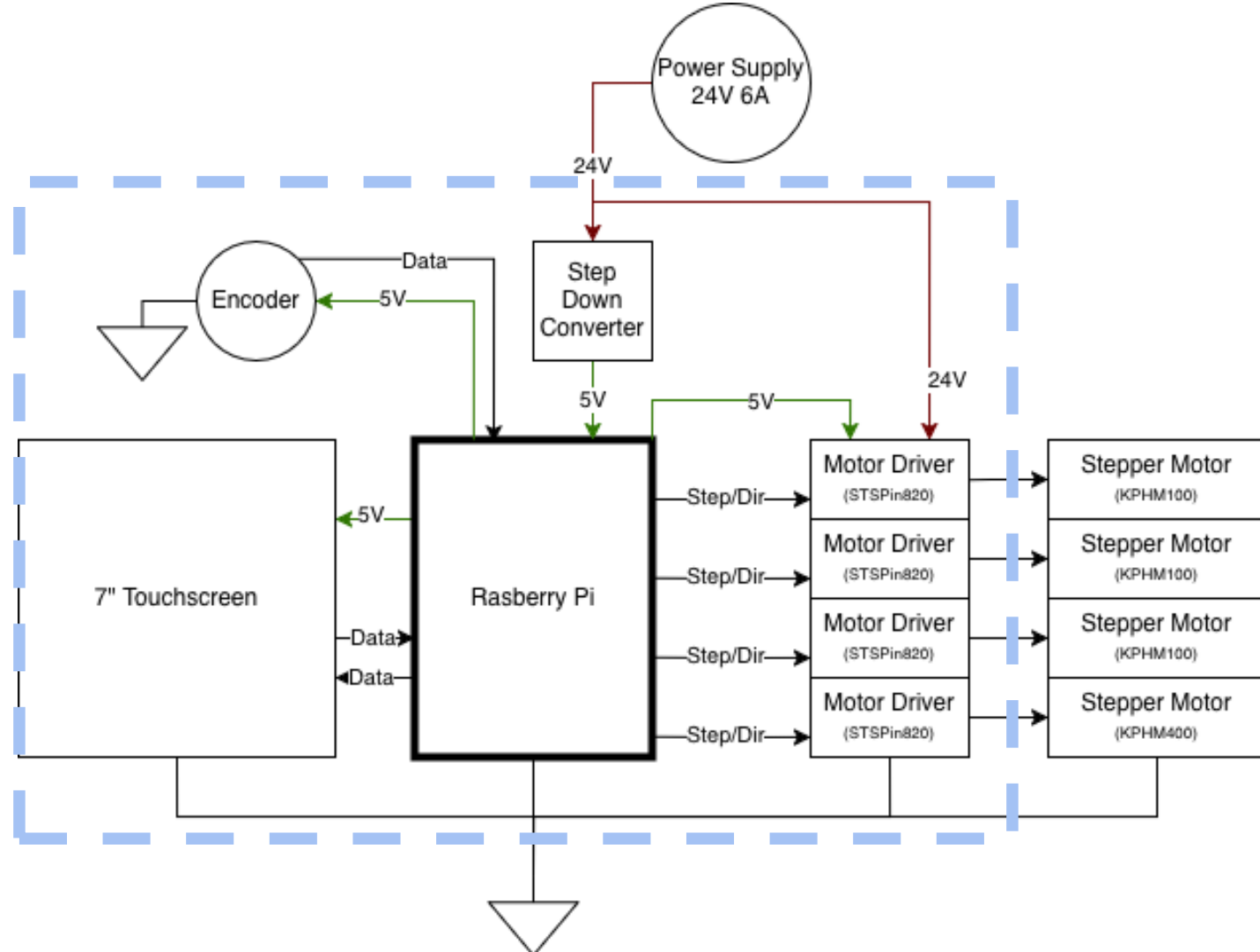
80

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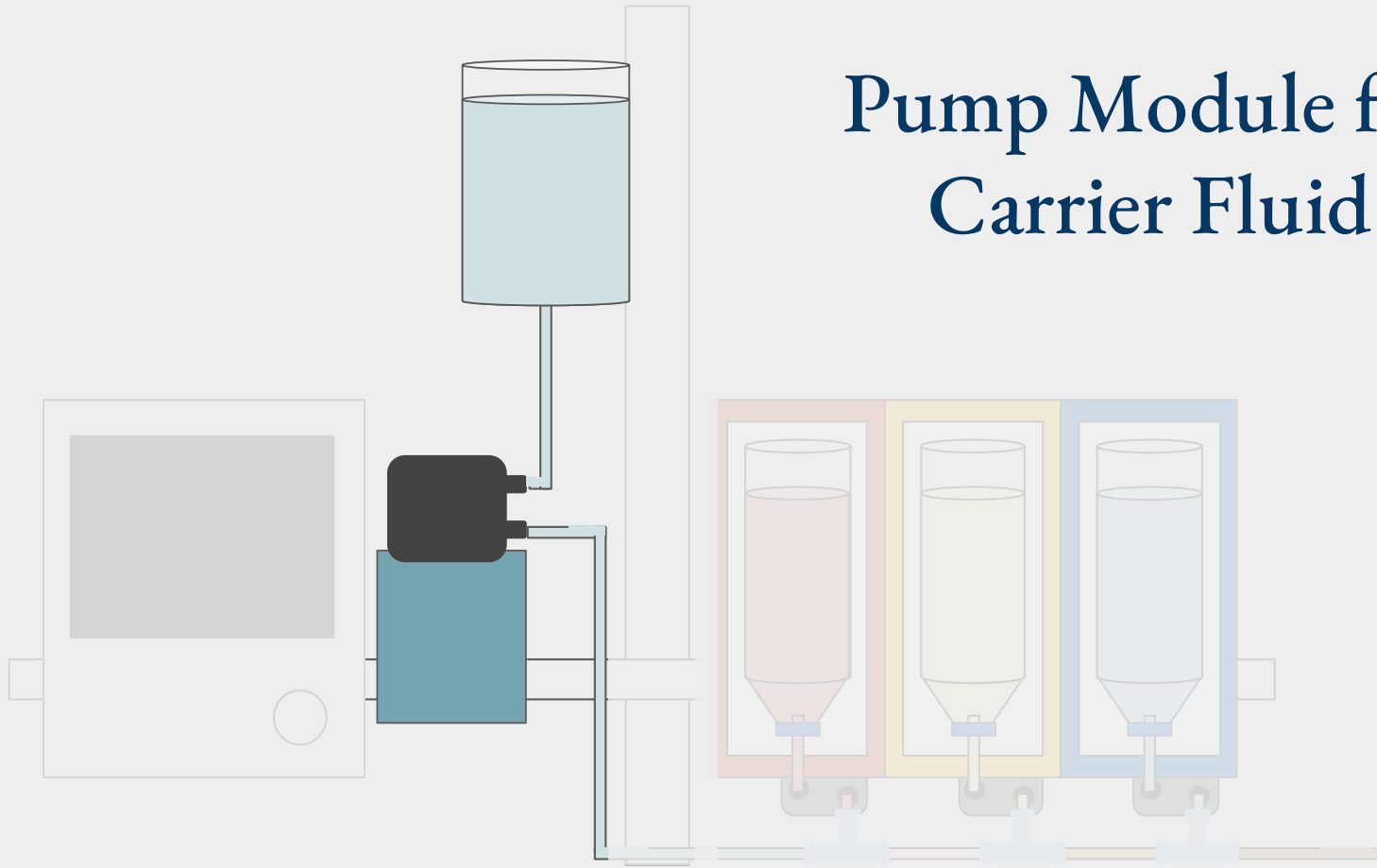
Custom

■ PAUSE

Stop All



Pump Module for Carrier Fluid



Pump Module for Carrier Fluid

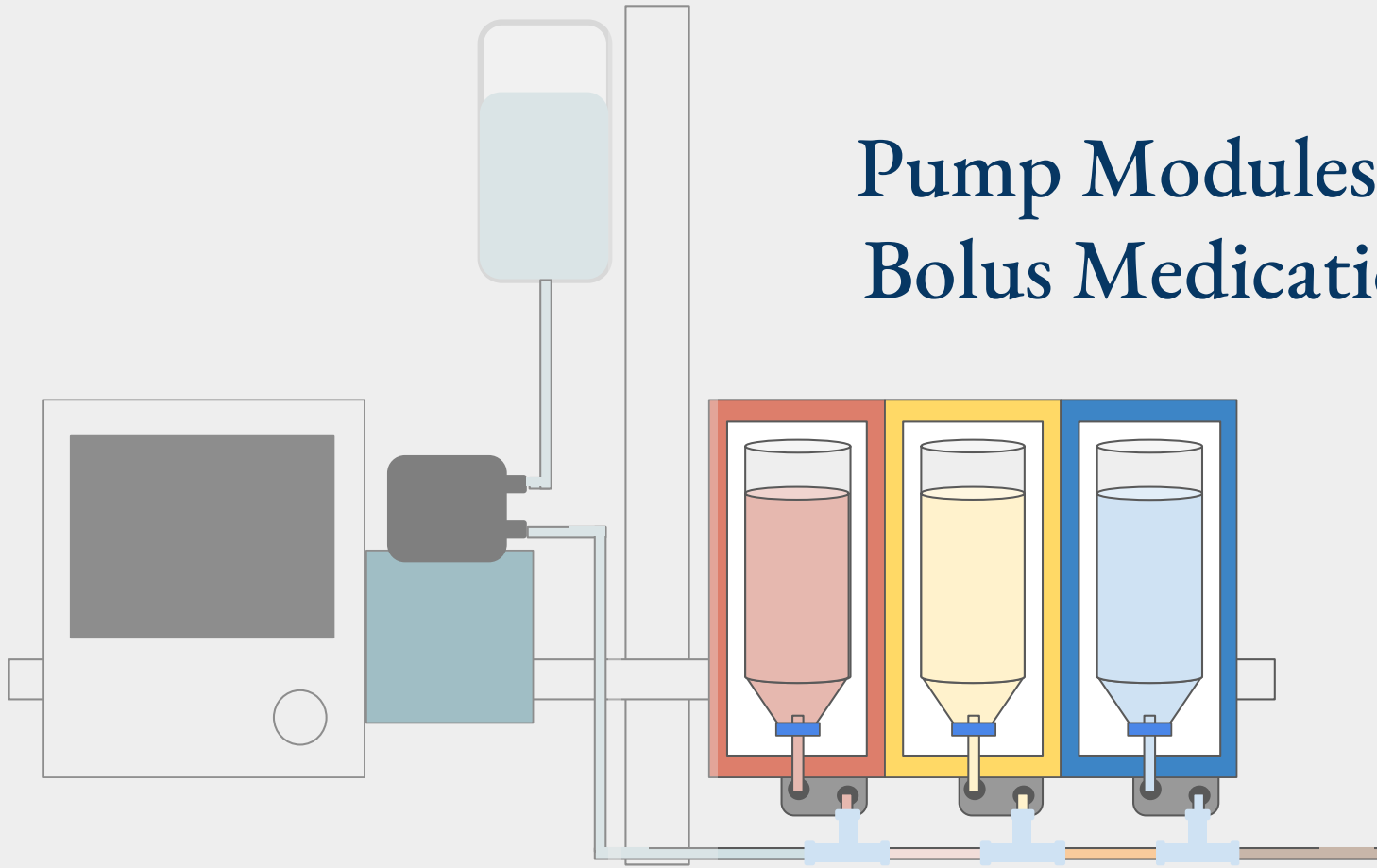


Pump Module for Carrier Fluid

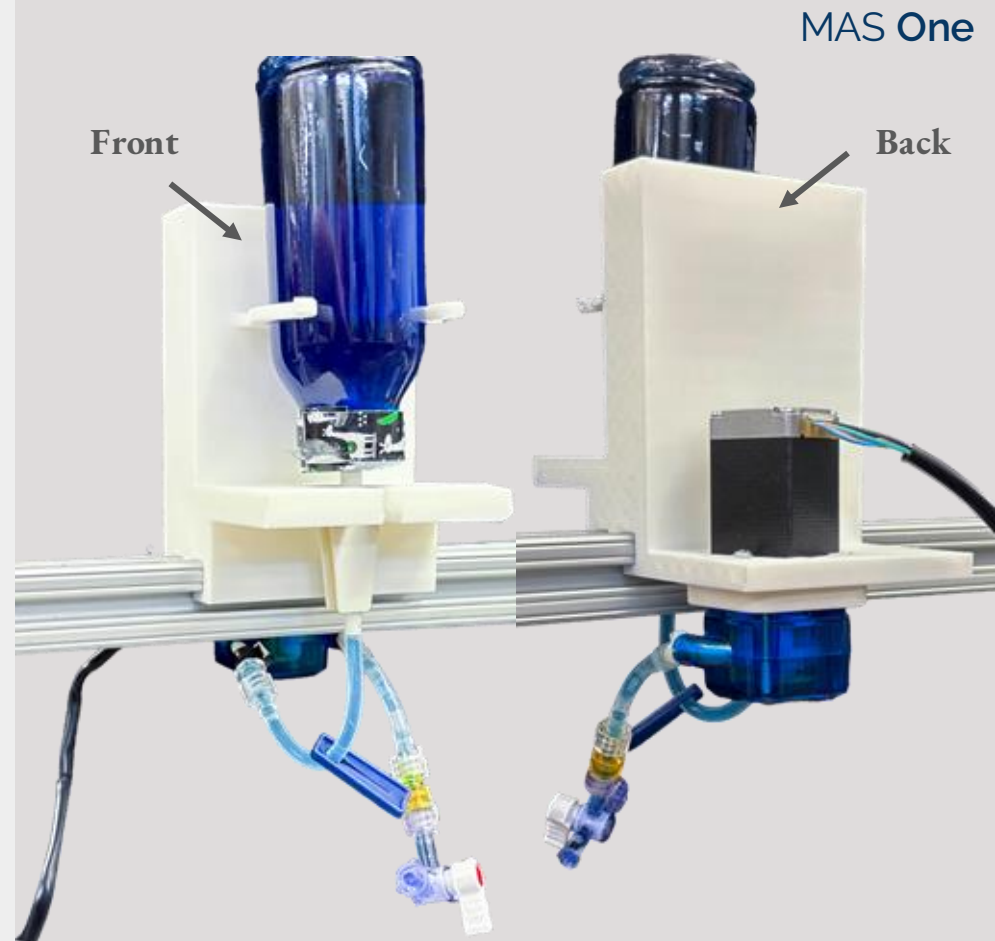
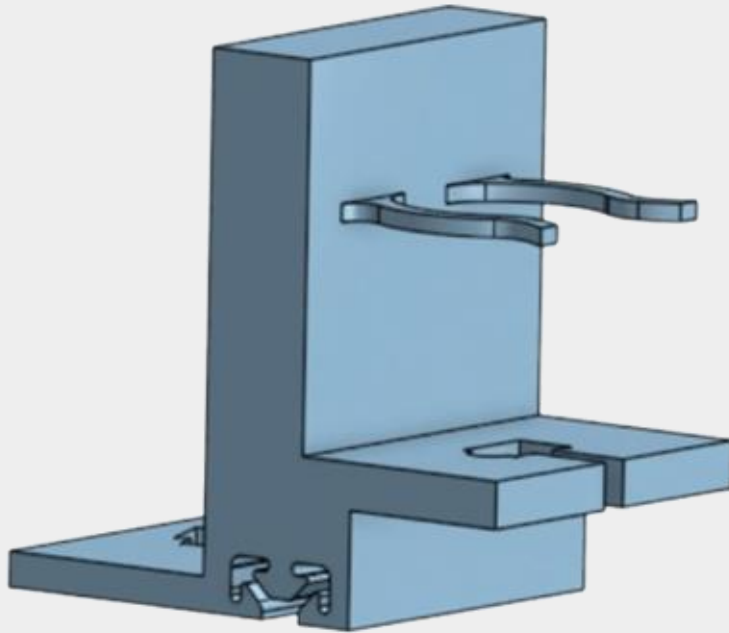
- Pumps saline carrier fluid from typical IV bag into the IV tubing, pushing the bolus medications into the patient as needed
- Runs at a consistent, programmable infusion rate up to 400mL/min, or enough to push medications through the line in 0.5 seconds
- Housing holds pump and motor, connecting them to the rail



Pump Modules for Bolus Medications

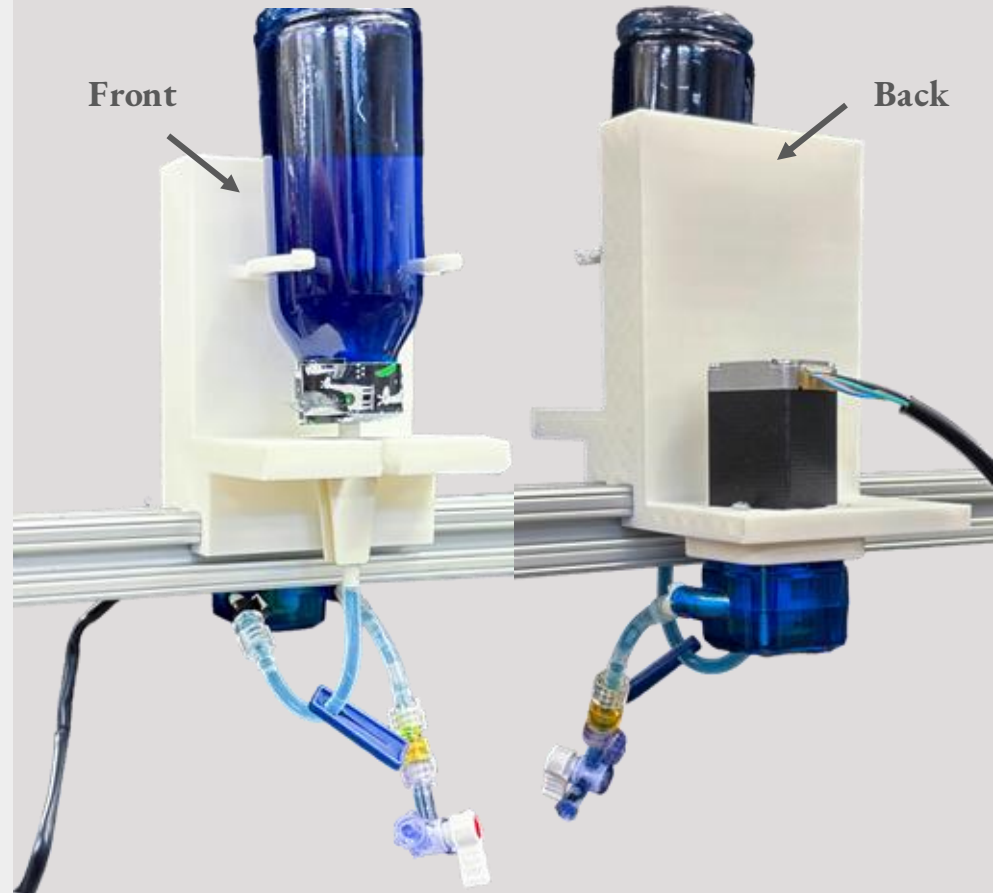


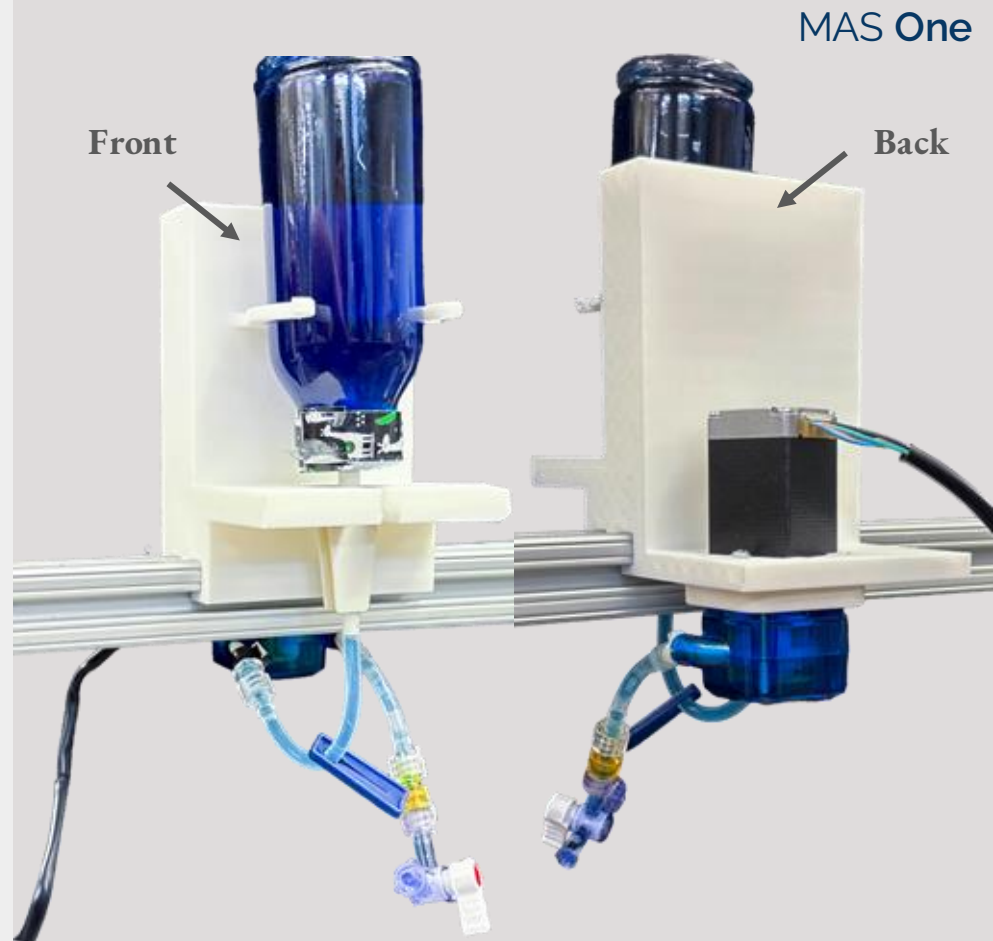
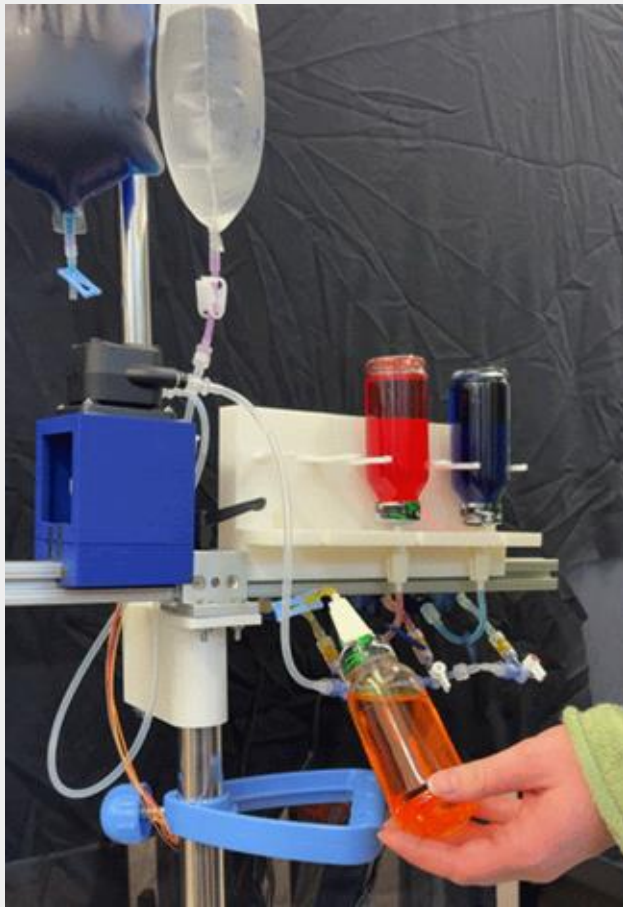
Pump Module for Bolus Medications



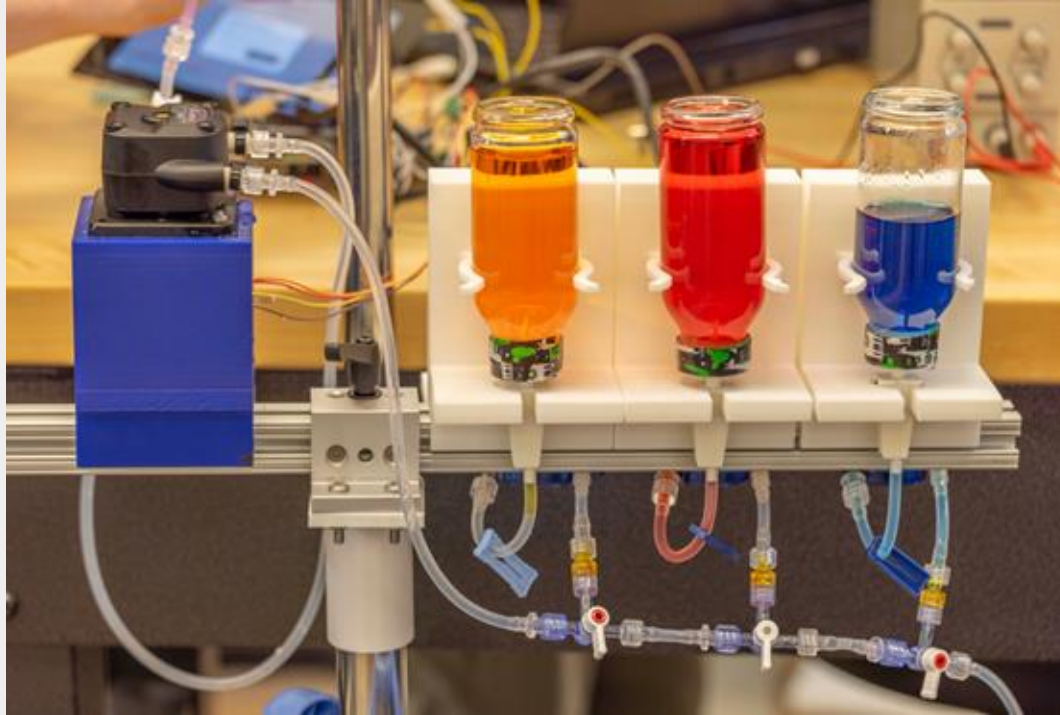
Pump Module for Bolus Medications

- Bolus modules capable of 0.1 cc resolution
- Designed to hold large medication vials for all-day use
- Holds vented spike with tubing connected to motor
- Peristaltic pumps mounted at the bottom with motors in the back
- One-way valve included after each pump to prevent inter-medication mixing





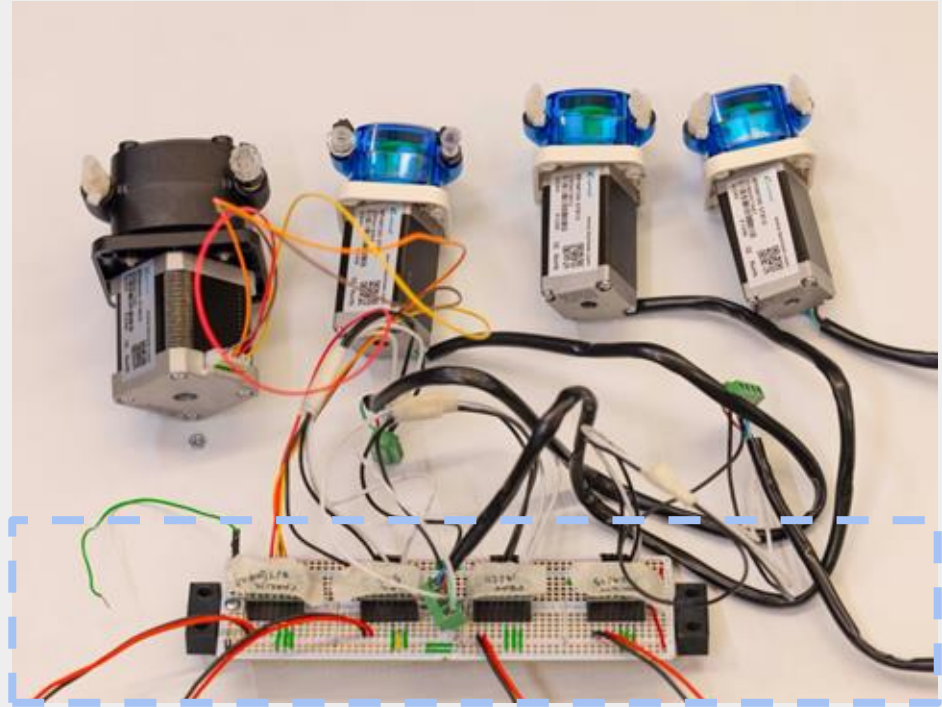
Pump Modules

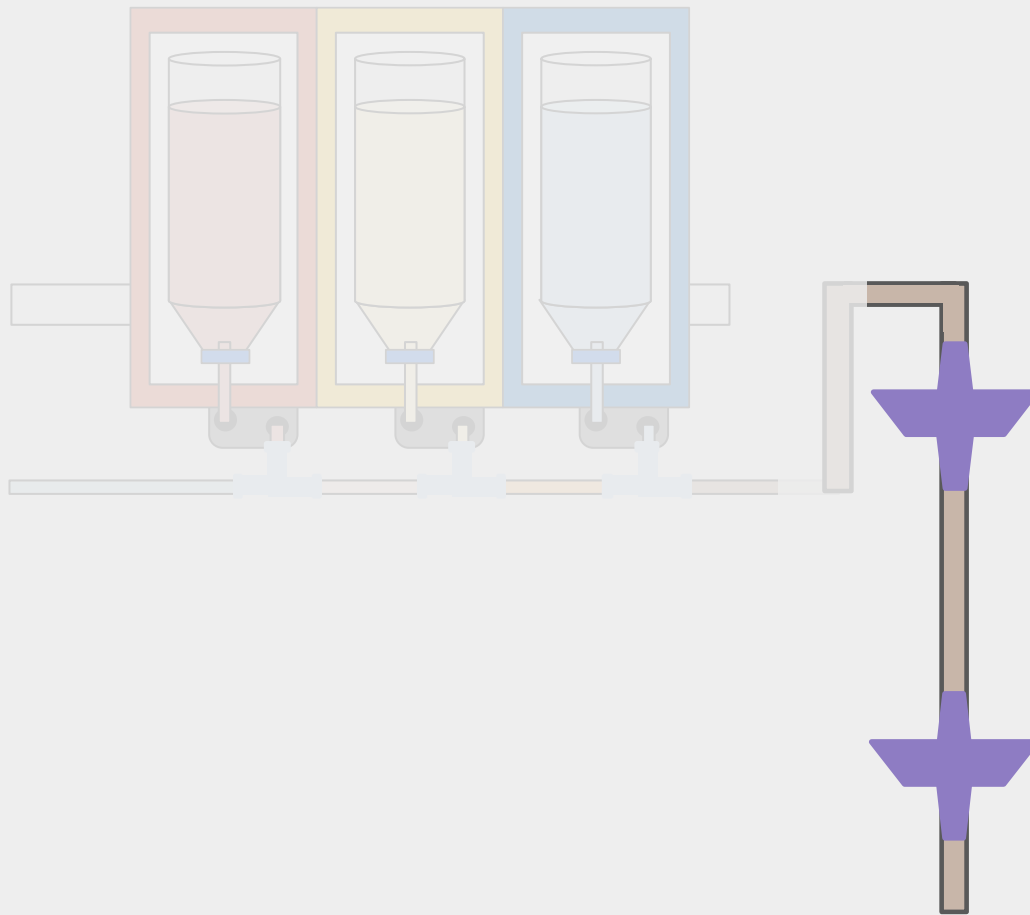


Pump Modules - Electrical Design

- Modules contain pump but not supporting drivers
- Carrier fluid and bolus pumps are peristaltic pumps powered by stepper motors
- Each plugs into the driver board (highlighted in blue) which is included in the central interface

Motor Model	Max Flow Rate	Resolution
KPHM100	65 mL/min	0.26 mL per rev
KPHM400	400 mL/min	1 mL per rev

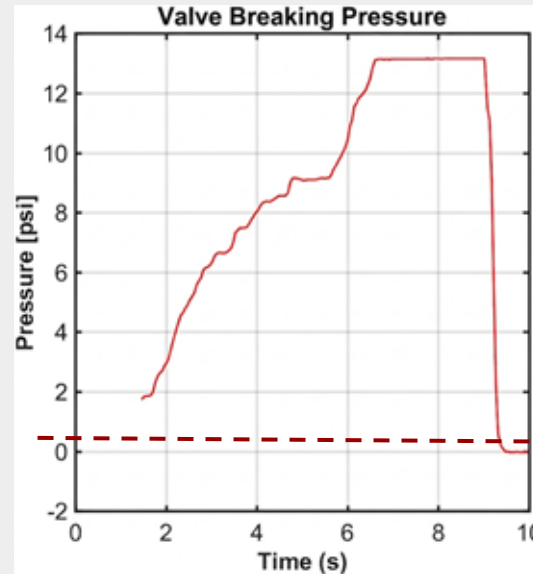
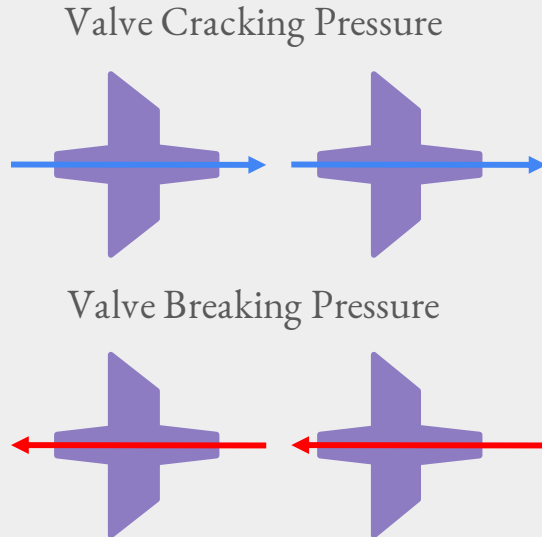




Contamination Prevention Module

Contamination Prevention Module - Mechanical Design

Two one way valves in series with cracking pressures of 5 psi and a breaking pressure >13 psi.



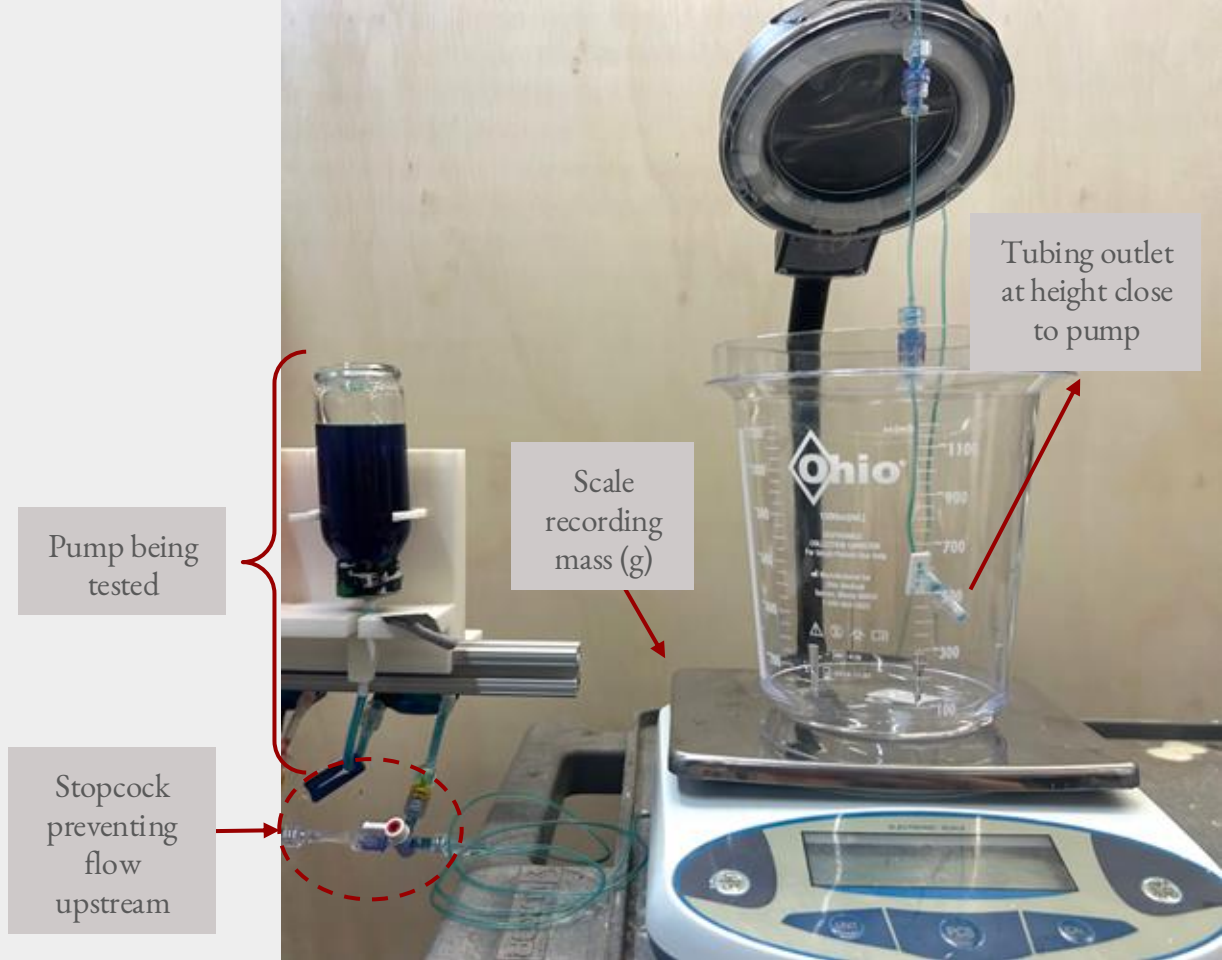
Typical venous blood pressure is only 0.04-0.29 psi



Testing and Verification

Bolus Dose Testing

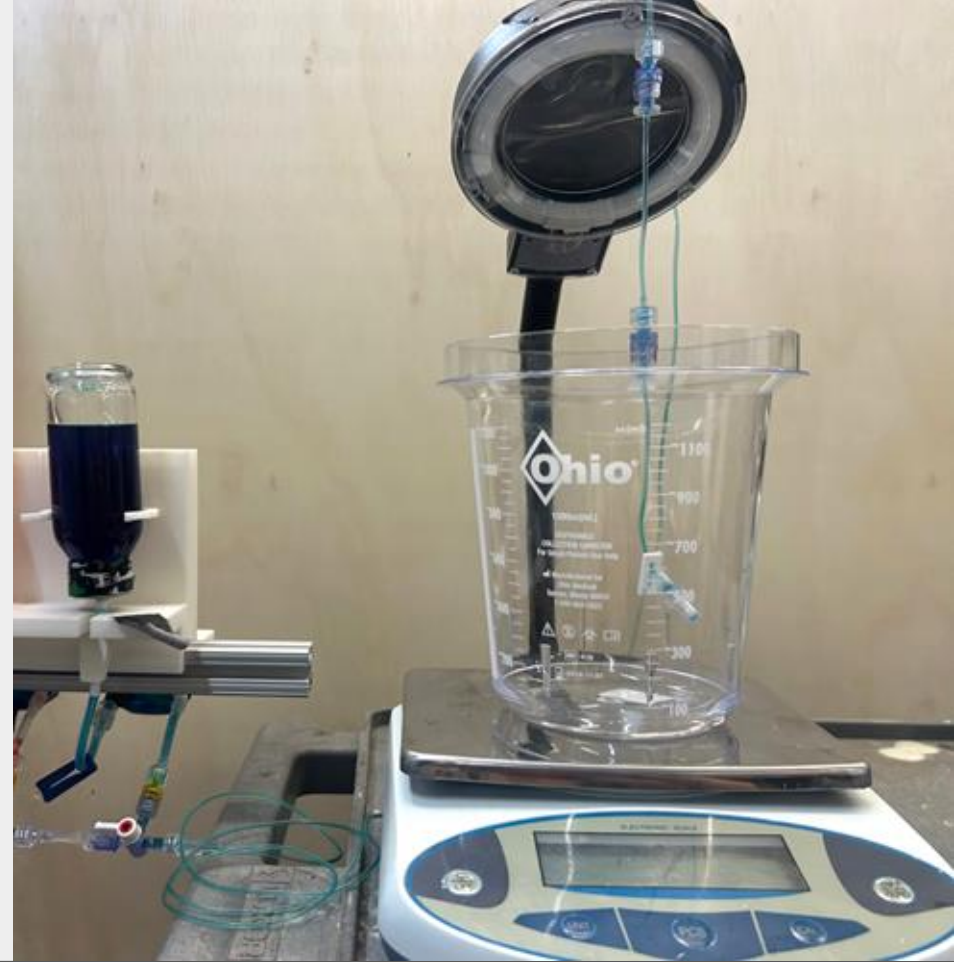
- Modified version of Accuracy test for pumps type 3* in IEC 60601-2-24
- Without carrier pump running, bolus dose into a primed IV line (with the contamination prevention module attached) and weigh the output
- 25 trials each of minimum 0.1cc and maximum 10cc bolus dose



Bolus Dose Testing Results

	Volume (cc)	Mean (cc)	Standard Deviation (cc)	Percent Deviation
Min Dose	0.10	0.10	0.01*	10.00%
Max Dose	10.00	10.00	0.02	0.20%

*Note the resolution of the scale used was 0.01 g (0.01 cc of water)



Design Evaluation and Further Work

Functional Requirement

Evaluation

Reduced volume of waste generated



Yes- replaces small volume syringes with large volume vials and day-use tubing

Minimized setup time and medication change time



Yes- requires only one section of tubing to be replaced between surgeries

Variable dosing of medications with 0.1cc resolution



Yes- minimum dose of 0.1 cc with 0.01 cc standard deviation

Simultaneous control of multiple medications



Yes- multiple medications controlled from one interface

Meet 5 second medicine administration window



Yes- Fast pumps and tubing volume of 3 mL means medicine can reach the patient in as little as 0.5 s

Compliance with relevant standards for safety/sterility



Further testing needed on safety of tubing changes and effectiveness of one-way valves for contamination prevention

Future Work

- Advanced functionality:
 - Pre-programmed drug delivery “macros”
 - Connection to electronic health records
 - Recording medication administered during procedure to streamline charting
- More user testing to inform improvements to the UI
- Development and extensive testing of contamination prevention module
- Wireless control system

Acknowledgements

Thank you to our project sponsor Neil Rens, our team mentors Anthony Pennes and Nevan Hanumara, our communication instructor Kate Parsons, and Jay Connor for his lovely insights and photos. Another special thank you to Dr. Paul H. Alfille and the MGH Department of Anesthesiology for giving us the opportunity to shadow in the OR.

Thank you!
Questions?

